25900354 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECT201, Discipline Specific Elective (DSE)-MG3DSEECC200

PROGRAMMING IN C

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. The increment/decrement part in a for loop [K] / [CO1]

   a). must be executed once before the loop starts

   b). executes at the end of each iteration

   c). is optional

   d). both b and c

2. Which of the following describes a function prototype in C? [U] / [CO2]

   a). It defines the body of the function

   b). It declares the function to the compiler before use

   c). It executes the function

   d). It automatically includes header files

3. What happens if you partially initialize a structure? [K] / [CO3]

   a). Compiler error

   b). Remaining members are set to 0 (zero-initialized)

   c). Remaining members get garbage values

   d). All members must be initialized explicitly

4. What is the purpose of problem analysis in programming? [U] / [CO1]

a). To translate code into machine language
b). To identify the inputs, processes, and outputs
c). To remove errors from a program
d). To design flow charts only

5. What is the index of the first element in a C array? [K] / [CO2]

a). 1 b). 0 c). -1 d). Depends on the array declaration

6. What is the correct syntax to end a structure definition in C? [K] / [CO3]

a). with a semicolon (`;`) b). with a colon (`:`) c). with a dot (`.`) d). with a curly brace (`}`) only

7. Which control structure is most suitable for implementing a menu-driven program [K] / [CO1]

a). simple if
b). if-else
c). switch case
d). else if ladder

8. In C, the default return type of a function is: [K] / [CO2]

a). void
b). int
c). float
d). char

9. Using pointers, arrays and strings in C can be: [U] / [CO3]

a). Ignored completely
b). Accessed and manipulated more efficiently
c). Made error free automatically
d). Converted into machine instructions

10. After declaring `struct Point p;`, what will be the initial value of its members `x` and `y`? [U] / [CO3]

a). 0 b). NULL c). an error d). random garbage values

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. List three arithmetic operators in C and their functions. [U] / [CO1]

12. What happens if a local and global variable share the same name? Illustrate with an example. [K] / [CO2]

13. Write a note on declaring pointer variables. [K] / [CO3]

14. Write about the role of header files such as `<stdio.h>` and `<math.h>` in C programming. [U] / [CO1]

15. Explain pointer initialisation in C with suitable examples. [U] / [CO3]

16. Write a program using continue to print only odd numbers between 1 and 20. [U] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Define C tokens. Explain in detail the different types of tokens in C with examples. [U] / [CO1]

18. Write a C program to copy one string to another using the strcpy() function. [A] / [CO2]

19. Write a C program to demonstrate the use of function pointer and explain step by step. [A] / [CO3]

20. Write a C program to check whether a given string is a palindrome or not. [A] / [CO2]

[10x2 = 20]